

Complex L_p -Intersection Bodies: Uniform Convexity Near Hermitian Ellipsoids

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Lane 0 — candidate partial result

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Status. Partial answer to the open convexity question for $-2 < p < -1$. For each complex dimension, a single C^2 neighborhood of the Hermitian ellipsoids works simultaneously for every exponent in the missing interval and every admissible planar body C . The proof is qualitative but uniform. It does not settle arbitrary balanced convex bodies.

1 The source question

Ellmeyer–Hofstätter define complex L_p -intersection bodies by

$$\rho_{I_{C,p}K}(u)^{-p} = \int_K h_{Cu}(x)^p dx, \quad -2 < p < 0,$$

where $C \subset \mathbb{C}$ contains the origin in its interior. For an $\mathbb{S}^1$ -invariant convex $K \subset \mathbb{C}^n$, they prove convexity when $p \geq -1$ and pseudoconvexity when $-2 < p < -1$, and explicitly ask whether the latter conclusion can be strengthened.

Theorem 1.2 ([5]). *Let $p > -1$ be non-zero. If $K \in \mathcal{S}_0(\mathbb{R}^n)$ is convex and origin-symmetric, then $I_p K$ is convex.*

Here, the condition that the body K is origin-symmetric, that is, K is the unit ball of a (real) norm cannot be omitted. Indeed, by translating the convex body K one can show that the intersection body of a translate is not convex anymore (see [10, Thm. 8.1.8] and [6]). Transferring the symmetry condition to the complex setting, real norms are naturally substituted by complex norms, that is, origin-symmetry is replaced by $\mathbb{S}^1$ -invariance, leading to the complex convexity theorem in [30] for I_c .

Theorem 1.3 ([30]). *If $K \in \mathcal{S}_0(\mathbb{C}^n)$ is convex and $\mathbb{S}^1$ -invariant, then $I_c K$ is convex.*

It is a natural question to ask whether complex L_p -intersection bodies are convex. As our second main result, we extend Theorems 1.2 and 1.3 to complex L_p -intersection bodies of $\mathbb{S}^1$ -invariant convex bodies in $\mathbb{C}^n$, weakening for $-2 < p < -1$ convexity to pseudo-convexity (see Section 4 for the definition). It is an interesting (open) question whether pseudo-convexity can be strengthened to convexity.

Theorem B. *Suppose that $C \in \mathcal{K}_{(0)}(\mathbb{C})$. If $K \in \mathcal{S}_0(\mathbb{C}^n)$ is convex and $\mathbb{S}^1$ -invariant, then $I_{C,p} K$ is pseudo-convex, if $-2 < p < -1$, and $I_{C,p} K$ is convex, if $-1 \leq p \neq 0$.*

The proof of Theorem B is very much inspired by the techniques from [5] and relies for $p > -1$ on Theorem 1.2. Indeed, for $p > -1$, we actually show, using techniques from isometric embeddings into L_p -spaces, the following close relation.

Theorem C. *Suppose that $p > -1$ is non-zero and $C \in \mathcal{K}_{(0)}(\mathbb{C})$. Then there exists $d_{C,p} > 0$, such that $I_{C,p} K = d_{C,p} I_p K$ for every $\mathbb{S}^1$ -invariant $K \in \mathcal{S}_0(\mathbb{C}^n)$.*

Let us also note that, in general, Theorem B without the assumption of $\mathbb{S}^1$ -invariance is false, if $p \geq -1$, as we show in Section 4.4.

Turning now to inequalities for intersection bodies, our next main result relates the volume of complex L_p -intersection bodies with the volume of their real counterparts.

Theorem D. *Suppose that $C \in \mathcal{K}_{(0)}(\mathbb{C})$ is origin-symmetric and $-1 \leq p < 1$ is non-zero. If $K \in \mathcal{S}_0(\mathbb{C}^n)$, then*

$$(7) \quad V_{2n}(I_{C,p} K) / V_{2n}(I_{C,p} B^{2n}) \leq V_{2n}(I_p K) / V_{2n}(I_p B^{2n}).$$

Let us note that the equality cases of (7) can be completely described by a technical statement in terms of the convex body C and will be stated later in Section 5. By Theorem C clearly $\mathbb{S}^1$ -invariant bodies satisfy equality. From Theorem D, we deduce the following generalization of Busemann's intersection inequality [2] for $I_{C,p}$ leading to affine isoperimetric inequalities in the following sense. Here, we call an ellipsoid E Hermitian, if $E = \varphi(B^{2n}) + t$ for $\varphi \in \text{GL}(n, \mathbb{C}), t \in \mathbb{C}^n$.

Corollary E. *Suppose that $C \in \mathcal{K}_{(0)}(\mathbb{C})$ is origin symmetric and $0 < p < 1$ or $-1 \leq p < 0$ and $n/|p| \in \mathbb{N}$. Among $K \in \mathcal{S}_0(\mathbb{C}^n)$, the ratio*

$$V_{2n}(I_{C,p} K) / V_{2n}(K)^{2n+p}$$

is maximized by origin-symmetric Hermitian ellipsoids. If $p = -1$, these are the only maximizers.

Figure 1: Compiled source page 4: the open convexity question and Theorem B.

For balanced data the source also proves that replacing C by the unit disk only changes the output by a homothety. In the disk case, putting $q = -p$, the reciprocal radial function is, up to a constant,

$$N_q(u) = \left(\int_K |x \cdot u|^{-q} dx \right)^{-1/q}, \quad 1 < q < 2. \quad (1.1)$$

Thus the full question asks whether (1.1) is always a complex norm.

2 Statement of the partial result

Use the covariant C^2 norm on the unit sphere. A body is C^2 -close to a Hermitian ellipsoid if some complex linear image has a C^2 radial function close to one.

Theorem 1 (one neighborhood for the whole missing interval). *For every $n \geq 2$ there is $\varepsilon_n > 0$ with the following property. Let $K \subset \mathbb{C}^n$ be an $\mathbb{S}^1$ -invariant convex body with a C^2 radial function. Suppose that for some $\Theta \in \mathrm{GL}(n, \mathbb{C})$,*

$$\|\rho_{\Theta K} - 1\|_{C^2(\mathbb{S}^{2n-1})} < \varepsilon_n. \quad (2.1)$$

Then $I_{C,p}K$ is a strictly convex body for every

$$C \in \mathcal{K}_{(0)}(\mathbb{C}), \quad -2 < p < -1. \quad (2.2)$$

The same ε_n works simultaneously for all p and all C in (2.2).

The input convexity assumption is included to match the source question. In fact, the proof only needs a positive balanced C^2 radial function satisfying (2.1).

3 A C^2 endpoint lemma

For $p > -2$ and $f \in C(\mathbb{S}^{2n-1})$, set

$$(T_p f)(u) = \frac{1}{\Gamma(p+2)} \int_{\mathbb{S}^{2n-1}} |v \cdot u|^p f(v) dv. \quad (3.1)$$

At $p = -2$, define $T_{-2} = c_n \mathcal{R}_c$, where $\mathcal{R}_c f(u)$ is the integral of f over the complex equator $\{v : v \cdot u = 0\}$ and $c_n > 0$ is the endpoint constant. Its exact value will not matter.

Lemma 2 (C^2 strengthening of endpoint convergence). *The map*

$$[-2, -1] \times C^2(\mathbb{S}^{2n-1}) \longrightarrow C^2(\mathbb{S}^{2n-1}), \quad (p, f) \longmapsto T_p f, \quad (3.2)$$

is continuous.

Proof. The source proves the C^0 convergence $T_p f \rightarrow c_n \mathcal{R}_c f$ as $p \downarrow -2$, including the version in which $f = f_p$ converges uniformly. Away from -2 , joint continuity follows directly from the locally integrable kernel.

It remains to retain two derivatives at the endpoint. The operators T_p are $\mathrm{U}(n)$ -equivariant. If X is an infinitesimal unitary rotation, differentiating the equivariance identity gives

$$X(T_p f) = T_p(Xf), \quad XY(T_p f) = T_p(XYf). \quad (3.3)$$

Choose finitely many such rotation fields which span every tangent space of the sphere. The C^2 norm is equivalent to the supremum norm of the function and of the resulting first and second derivatives. Apply the source's C^0 endpoint convergence separately to f , Xf , and XYf . Equation (3.3) then gives convergence in C^2 . The same argument with $f_p \rightarrow f$ in C^2 proves joint continuity at $p = -2$. $\square$

Remark 3. *The normalization in (3.1) is essential: $\Gamma(p+2)$ removes the mass blow-up of $|v \cdot u|^p$ along the complex equator. The limiting statement is the operator form of the source's interpolation between complex L_p -intersection bodies and the complex intersection body.*

4 Proof of Theorem 1

The source's $\mathrm{GL}(n, \mathbb{C})$ contravariance shows that a complex linear change of the input produces a complex linear change and a positive scaling of the output. Convexity is unchanged, so assume Θ in (2.1) is the identity. On balanced functions, angular averaging in C reduces $\mathcal{I}_{C,p}$ to $\mathcal{I}_{\mathbb{D},p}$ up to a positive scalar. It is therefore enough to treat $C = \mathbb{D}$.

For $p > -2$, polar coordinates give

$$\rho_{\mathbb{I}_{\mathbb{D},p}K}(u)^{-p} = \frac{1}{2n+p} \mathcal{J}_{\mathbb{D},p}(\rho_K^{2n+p})(u). \quad (4.1)$$

Positive scalings do not affect convexity. Divide (4.1) by $\Gamma(p+2)$ and by its value for the Euclidean ball, and define on the sphere

$$g_{p,K}(u) = \left(\frac{T_p(\rho_K^{2n+p})(u)}{T_p 1} \right)^{1/p}, \quad -2 < p \leq -1. \quad (4.2)$$

This is the reciprocal radial function of a homothetic copy of $\mathbb{I}_{\mathbb{D},p}K$. At $p = -2$, use the continuous definition

$$g_{-2,K}(u) = \left(\frac{T_{-2}(\rho_K^{2n-2})(u)}{T_{-2} 1} \right)^{-1/2}. \quad (4.3)$$

By Lemma 2, the elementary continuity of $(p, r) \mapsto r^{2n+p}$ in C^2 near $r = 1$, and positivity of the denominator,

$$(p, \rho_K) \mapsto g_{p,K} \text{ is continuous from } [-2, -1] \times \mathcal{U} \longrightarrow C^2(\mathbb{S}^{2n-1}), \quad (4.4)$$

where $\mathcal{U}$ is a sufficiently small C^2 neighborhood of one. For the ball, $g_{p,B} = 1$ for every $p \in [-2, -1]$.

Compactness of $[-2, -1]$ and (4.4) now provide one $\varepsilon_n > 0$ such that

$$\|\rho_K - 1\|_{C^2} < \varepsilon_n \implies \sup_{-2 \leq p \leq -1} \|g_{p,K} - 1\|_{C^2} < \frac{1}{4}. \quad (4.5)$$

Shrink ε_n if necessary to absorb the fixed equivalence constants in the chosen C^2 norm.

Recall the spherical curvature criterion: the one-homogeneous extension of a positive $g \in C^2(\mathbb{S}^{2n-1})$ is a strictly convex gauge when

$$Q_g := \nabla_{\mathbb{S}}^2 g + g \text{Id} > 0 \quad (4.6)$$

on every tangent space. At $g = 1$, $Q_g = \text{Id}$. Estimate (4.5) therefore implies $Q_{g_{p,K}} \geq \frac{1}{2} \text{Id}$ after another harmless reduction of ε_n . Hence the body with gauge $g_{p,K}$ is strictly convex for all $p \in [-2, -1]$. Undoing the homotheties, the reduction of C , and the complex linear map proves the theorem. $\square$

5 Why this does not continue to every balanced body

One might join the ball to an arbitrary balanced convex body and try an open–closed argument. The proof above makes convexity open as long as the real curvature matrix remains positive, and convexity is closed under radial limits. The missing step is rigidity at a first zero curvature.

This is precisely where pseudoconvexity is weaker. In a complex tangent line it controls the Levi trace

$$D^2 g(v, v) + D^2 g(iv, iv), \quad (5.1)$$

whereas real convexity needs both summands separately nonnegative. At a first degeneracy one summand can in principle cross zero while the other remains positive. The section-moment maximum in the source proves the sign of (5.1) but supplies no equality rigidity for either summand. Thus the local theorem does not silently assume the missing global estimate.

Eight focused upgrade attempts were made: negative- L^p inequalities, explicit Reinhardt bodies, balanced concentration limits, real-curvature formulas, spherical-harmonic linearization, singularity-adapted $\mathbb{C}^2$ polytope searches, endpoint interpolation, and global deformation. The counterexample alarms from coarse numerical Hessians vanished under direct high-resolution midpoint tests. The endpoint argument above is the strongest rigorous conclusion obtained.

6 Novelty and verification scope

The proof uses only the source’s transform formula, reduction to the disk, $\mathrm{GL}(n, \mathbb{C})$ covariance, and normalized C^0 endpoint convergence, plus the standard spherical curvature criterion. The new step is to commute first and second rotation derivatives through the normalized transforms and then use compactness in p before choosing the neighborhood of the ellipsoid.

Searches through the source’s citations, exact phrases, later arXiv records, and the run indexes did not locate this simultaneous local theorem. Because the argument is a natural stability consequence of known endpoint machinery, it should be treated as a candidate partial result pending expert review, not as a claim that the global open problem is solved.

References

- [1] S. Ellmeyer and G. C. Hofstätter, *Complex L_p -intersection bodies*, Adv. Math. 431 (2023), 109247, [arXiv:2304.00794](#).
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- [3] G. Berck, *Convexity of L_p -intersection bodies*, Adv. Math. 222 (2009), 920–936.